Sales Performance Analysis

End-to-End Case Study: Where do we make money, where do we lose it, and what should we do?

Data: 700 orders · 5 segments · 5 countries · 6 products · Sep 2013 - Dec 2014

$118.7M	$16.9M	14.2%	1.13M	$9.2M	58
Net Sales	Profit	Profit Margin	Units Sold	Discounts Given	Loss-making Orders

Headline: Sales are growing fast (+36.9% like-for-like), but **discounting is eroding margin** and the **Enterprise segment loses money** (-$0.6M). Fixing discount discipline in Enterprise and High-discount deals is the single biggest profit lever.

Executive Summary

- **Overall:** $118.7M net sales, $16.9M profit, 14.2% margin. Like-for-like (Sep-Dec), sales grew **+36.9%** and profit **+40.1%** from 2013 to 2014.
- **Profit engine:** Government generates **67%** of all profit ($11.4M at a 21.7% margin). Small Business adds another 24.5%.
- **Profit leak:** Enterprise sells $19.6M but loses $615K. Its list price ($125) is only 4% above unit cost ($120), so any discount above 4% makes an order unprofitable. All 58 loss-making orders are Enterprise orders with Medium or High discounts.
- **Discount effect:** Margin falls from **21.9%** (no discount) to 17.9% (low), 14.4% (medium) and **9.1%** (high). The pattern holds in every segment.
- **Hidden opportunity:** Channel Partners earn a **73%** margin but are only 1.5% of sales; Midmarket earns 28% on 2% of sales.
- **Seasonality:** Sales peak in **October and December 2014** (about $12M each).

1. Objective, Data & Method

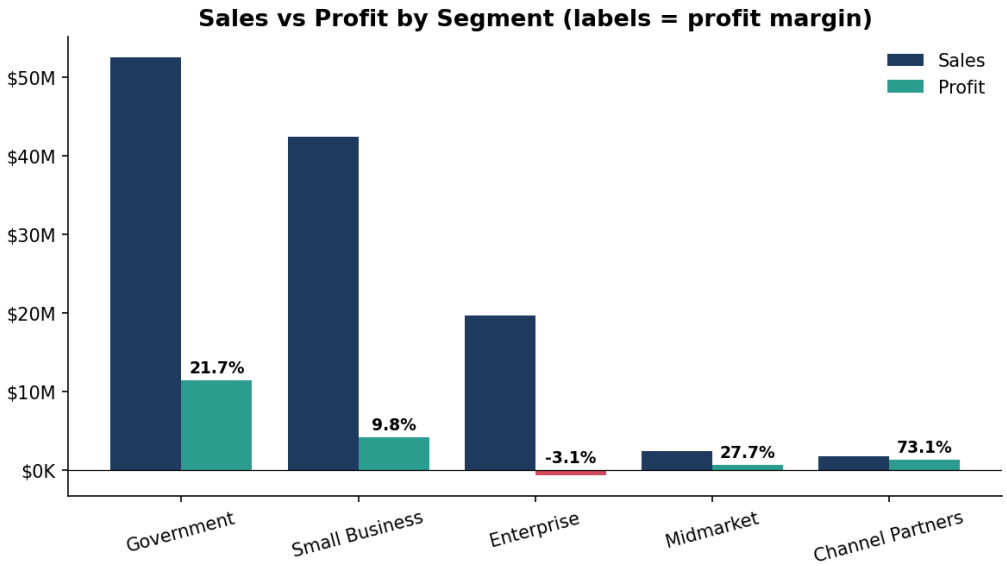
Objective: find what drives profit and loss in 16 months of sales and recommend actions. **Workflow:** Python (Pandas, Matplotlib, Seaborn) for cleaning/EDA → SQL (MySQL 8) for analysis → Excel and Power BI for dashboards → this report.

Data cleaning log

#	Issue found	Action taken
1	Column ' Sales' had a leading space	Headers trimmed; spaces replaced by underscores
2	Discount Band blank in 53 rows	All 53 rows have Discounts = 0, so they are no-discount sales: relabelled **No Discount** (not deleted)
3	36 rows have half-units (e.g. 4,492.5)	Kept as reported; flagged to the data owner
4	Manufacturing Price does not reconcile to COGS / Units (only 164 of 700 rows match)	COGS used as reported for profit; flagged as a data-governance issue
5	2013 contains only Sep-Dec; 2014 is a full year	Growth measured **like-for-like (Sep-Dec only)**. A full-year comparison would wrongly show +249%
6	Integrity checks	Gross - Discounts = Sales and Sales - COGS = Profit verified on all 700 rows; 0 duplicates; 0 nulls after cleaning

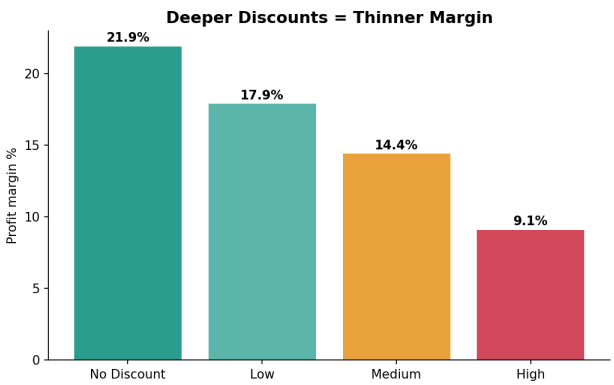
2. Findings

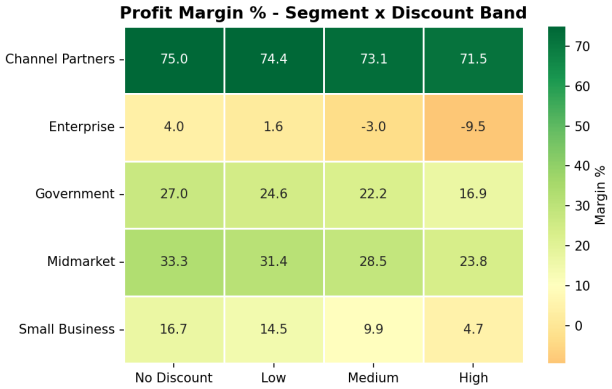
2.1 Segment performance: Government earns the profit, Enterprise loses it



- Government: $52.5M sales, $11.4M profit (21.7% margin) = 67% of total profit.
- Small Business: $42.4M sales but only 9.8% margin; the second-largest profit pool ($4.1M).
- Enterprise: $19.6M sales (3rd largest) but **-3.1% margin**.
- Channel Partners (73.1%) and Midmarket (27.7%) have the best margins but together are only 3.5% of sales.

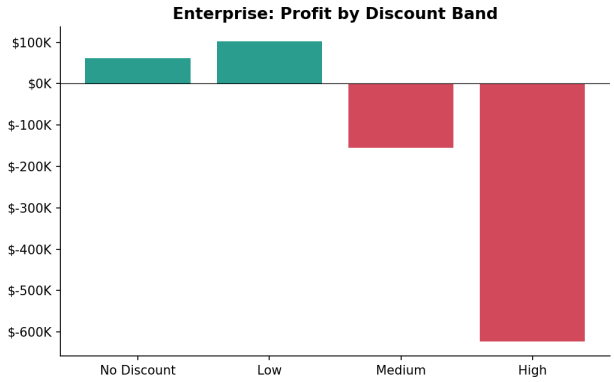
2.2 Discounts: deeper discount, thinner margin





- The average discount in the High band is 12.5% vs 2.5% in the Low band; margin drops from 17.9% to 9.1%.
- In the heatmap every segment gets worse as discounts deepen. Enterprise is the only one that turns negative (-3.0% Medium, -9.5% High).
- 245 orders (35%) are High-discount, so this is not an edge case.

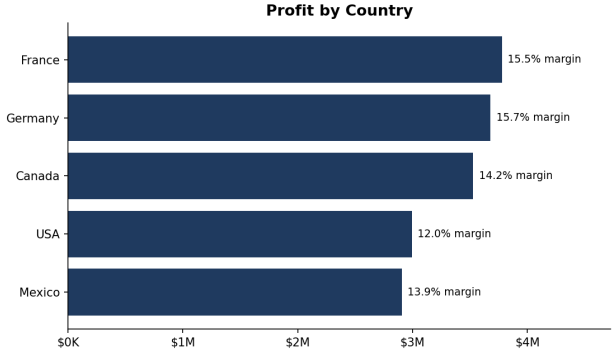
2.3 Enterprise deep-dive: a pricing problem, not a demand problem

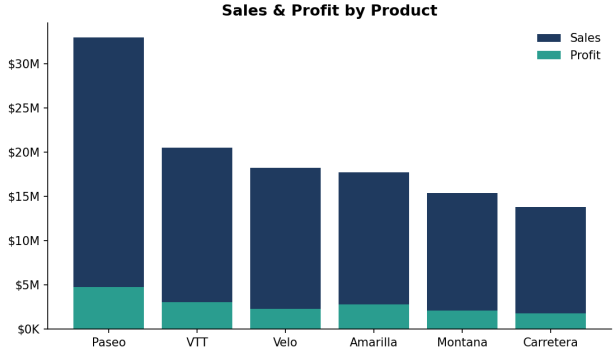


Enterprise metric	Value
List price / unit	$125
Cost (COGS) / unit	$120
Gross margin before discount	4.0%
Average discount given	6.9%
Actual profit	-$615K
Profit if no discounts*	+$843K

*Upper bound: assumes the same volume would have sold without discounts. Enterprise discounts total $1.46M, which is more than the entire loss.

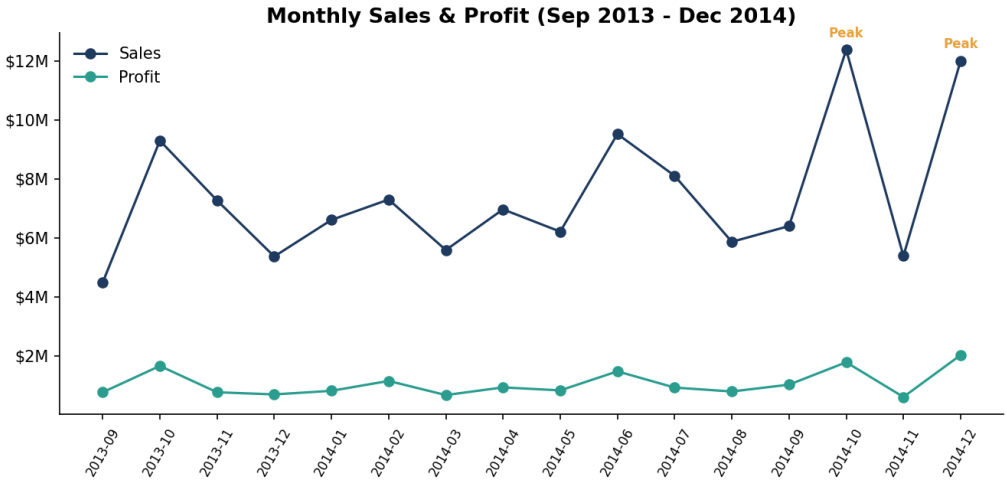
2.4 Country & product: balanced, so they are not the lever





- Country margins range 12.0% (USA) to 15.7% (Germany); the USA has the highest sales but the lowest margin.
- Product margins range 12.6% (Velo) to 15.9% (Amarilla). Paseo is the volume leader (30% of units) at a 14.5% margin.

2.5 Time trend: strong growth and Q4 peaks



- Like-for-like Sep-Dec: sales $26.4M → $36.2M (**+36.9%**); profit +40.1%, so growth did not come at the cost of margin.
- Peak months: Oct 2014 ($12.4M) and Dec 2014 ($12.0M). Nov 2014 dipped to $5.4M, so demand is lumpy.

3. Recommendations

#	Action	Evidence	Illustrative impact*
1	**Reprice or cap discounts in Enterprise** (max Low band, ~2.5%) or raise list price	4% pre-discount margin; all 58 loss orders are Enterprise Medium/High	+$0.98M profit; Enterprise from -$0.61M to +$362K
2	**Require approval for High-band discounts** (>10%) across all segments	High band margin 9.1% vs 17.9% Low; 245 orders	+$2.25M if High were capped at the Medium rate (7.2%)
3	**Grow Channel Partners and Midmarket** (partner programme, targeted sales)	73% and 28% margins on only 3.5% of sales	Every extra $1M of Channel sales adds about $0.73M profit
4	**Protect Government** (account management, renewals)	67% of profit from 300 orders	Risk reduction: a 10% Government dip costs about $1.1M
5	**Plan for Q4 peaks**; run promotions in low months instead of discounting at peak	Oct and Dec at about $12M vs $5-6M in other months	Better stock/staffing; fewer unnecessary discounts
6	**Fix data governance** (Manufacturing Price vs COGS, half-units)	Only 164 of 700 rows reconcile	More reliable margin reporting

*Priorities 1 and 2 overlap (Enterprise High orders appear in both) and must not be added together. All scenarios assume unchanged sales volume; real volume may fall if discounts are removed, so pilot before rolling out.

4. Limitations & Next Steps

- Only 16 months of data and a 4-month 2013, so seasonality is based on one full year.
- Profit uses COGS as reported; the Manufacturing Price column could not be reconciled.
- No customer-level or price-elasticity data, so discount scenarios are illustrative upper bounds.
- Next: test Enterprise pricing in one country for a quarter, build a discount-approval dashboard, and add customer IDs to the dataset for retention analysis.